Projekt: Gun Detector — Detekcja broni na zdjęciach

Cel: Zbudowanie detektora broni krótkiej i długiej na zdjęciach z użyciem modelu YOLOv8 (fine-tuning).

Dataset: Weapon-2 z Roboflow Universe (CC BY 4.0)

- 2 klasy: `guns` , `knife`
- Format: YOLO (bounding boxes)

Model: YOLOv8n (nano) — fine-tuning z pretrenowanych wag na 25 epokach

```
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
Device: cuda
PyTorch: 2.6.0+cu124
CUDA available: True
```

1. Konfiguracja datasetu

Eksploracja struktury datasetu, `data.yaml` , przykładowe anotacje i wizualizacja próbek treningowych z bounding boxami.

```
=== data.yaml ===
  train: train/images
  val: test/images
  test: test/images
  nc: 2
  names: ['guns', 'knife']
  roboflow: {'workspace': 'joao-assalim-xmovq', 'project': 'weapon-2', 'version': 2, 'lice
nse': 'CC BY 4.0', 'url': 'https://universe.roboflow.com/joao-assalim-xmovq/weapon-2/datas
et/2'}

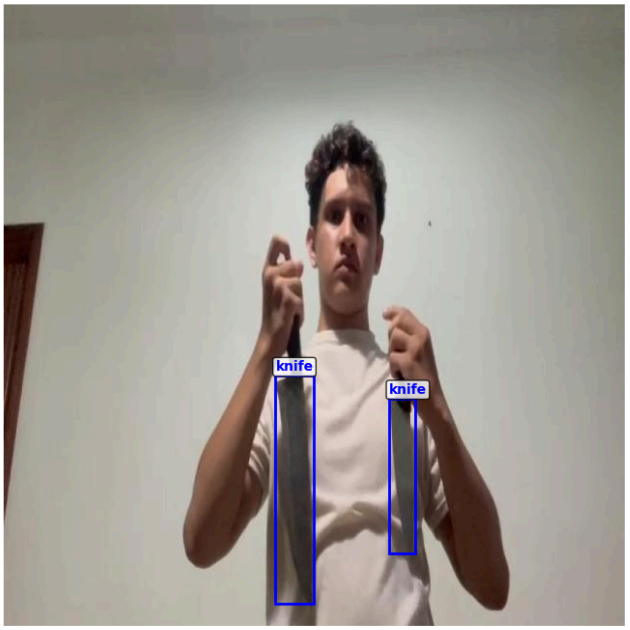
Struktura datasetu:
  Train: 3839 obrazów, 3839 etykiet
  Test:  259 obrazów, 259 etykiet

Klasy (2): ['guns', 'knife']

Przykładowa etykieta:
  Plik: Dikran_Tulaine-B92F_jpg.rf.9964ecb6403877b3743b521ca5c0953e.txt
  Zawartość: 0 0.490625 0.28984375 0.1515625 0.1609375
```

Przykładowe obrazy treningowe z anotacjami

WhatsApp-Video-2023-11-22-at-19_46_37_mp4-65_jpg.r



armas-2346-_jpg.rf.7212bb791576b71eeb9c75453370e11



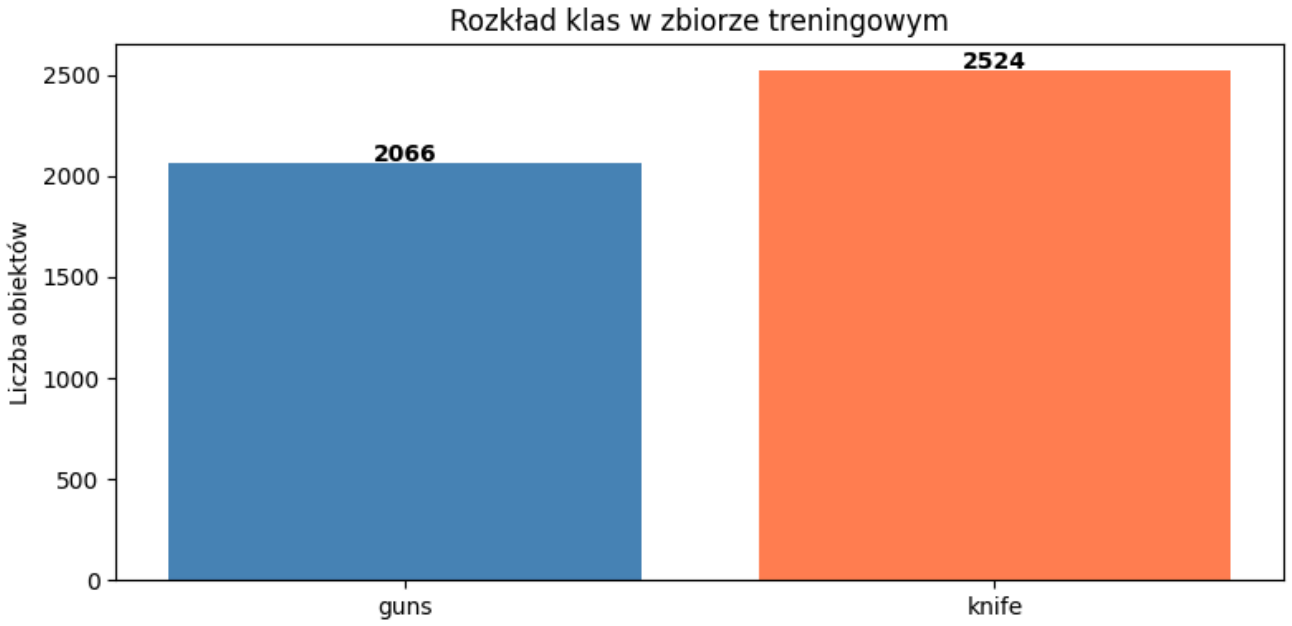
armas-232-_jpg.rf.beb982c9f002a19f8ce498a0097b00dd



armas-1708-_jpg.rf.6f6b1fe49582e107ff4d1f403f82d77




```
Rozkład klas w zbiorze treningowym:
  guns: 2066 obiektów
  knife: 2524 obiektów
```



2. Trening modelu YOLOv8

Fine-tuning modelu YOLOv8n (nano) na datasecie Weapon-2. Minimum 25 epok, z zapisem najlepszego modelu (`best.pt`).

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to
'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 22.3MB/s 0.3s.2s<0.1s.8s
New https://pypi.org/project/ultralytics/8.4.38 available  Update with 'pip install -U ult
ralytics'
Ultralytics 8.4.14  Python-3.12.6 torch-2.6.0+cu124 CUDA:0 (NVIDIA GeForce RTX 3060 Laptop
GPU, 6144MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randa
ugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10,
cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cut
mix=0.0, data=data\data.yaml, degrees=0.0, deterministic=True, device=None, dfl=1.5, dnn=F
alse, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=25, erasing=0.4, exist_
ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=Fals
e, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.
0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train,
model=yolov8n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=weapon_detector, nbs=6
4, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=10,
perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect
=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=F
alse, save_dir=C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\detect\w
eapon_detector, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=
0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, sim
plify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, ti
me=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.000
5, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=2

                   from  n    params  module                                       argumen
ts
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3, 16,
3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16, 3
2, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32, 3
2, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 6
4, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64, 6
4, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64, 12
8, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128, 1
28, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 2
56, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256, 2
56, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256, 2
56, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None,
2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384, 1
28, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None,
2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192, 6
4, 1]
```

```
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 6
4, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192, 1
28, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 1
28, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384, 2
56, 1]
 22        [15, 18, 21]  1    751702  ultralytics.nn.modules.head.Detect           [2, 16,
None, [64, 128, 256]]
Model summary: 130 layers, 3,011,238 parameters, 3,011,222 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'yolo26n.pt': 100% ━━━━━━━━━━━━ 5.3MB 20.1MB/s 0.3s.2s<0.1s.6s
AMP: checks passed
train: Fast image access  (ping: 0.20.0 ms, read: 2.81.3 MB/s, size: 29.0 KB)
train: Scanning C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\data\train\l
abels... 3839 images, 9 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3839/3839 518.9it/s 7.4s
0.1s
train: New cache created: C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\da
ta\train\labels.cache
val: Fast image access  (ping: 0.40.2 ms, read: 1.10.2 MB/s, size: 26.8 KB)
val: Scanning C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\data\test\labe
ls... 259 images, 5 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 259/259 84.3it/s 3.1s<0.0s
val: New cache created: C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\data
\test\labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determinin
g best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64
weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\dete
ct\weapon_detector\labels.jpg...
2026/04/17 08:59:18 INFO mlflow.tracking.fluent: Experiment with name '/Shared/Ultralytic
s' does not exist. Creating a new experiment.
```

```
MLflow: logging run_id(ff1b8f6c05ff490594cfe2246c4077eb) to runs\mlflow
MLflow: view at http://127.0.0.1:5000 with 'mlflow server --backend-store-uri runs\mlflow'
MLflow: disable with 'yolo settings mlflow=False'
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\detect\weapon_detector
Starting training for 25 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/25         2G      1.231      2.266      1.456         40        640: 100% ━━━━━━━━━━━━ 240/240 5.1it/s 47.0s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 4.5it/s 2.0s0.2s
                   all        259        481      0.808      0.369      0.387      0.253

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/25      2.32G        1.3      1.785      1.507         33        640: 100% ━━━━━━━━━━━━ 240/240 5.8it/s 41.6s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 5.0it/s 1.8s0.2s
                   all        259        481      0.431      0.402      0.375      0.229

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/25      2.32G      1.293      1.586      1.519         37        640: 100% ━━━━━━━━━━━━ 240/240 5.9it/s 40.6s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 5.0it/s 1.8s0.2s
                   all        259        481      0.785      0.411      0.532       0.34

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/25      2.32G      1.246      1.417      1.466         52        640: 100% ━━━━━━━━━━━━ 240/240 5.9it/s 40.4s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 4.8it/s 1.9s0.2s
                   all        259        481      0.743       0.53      0.594      0.397

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/25      2.33G       1.21      1.369      1.443         34        640: 100% ━━━━━━━━━━━━ 240/240 6.0it/s 40.3s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 5.2it/s 1.7s0.2s
                   all        259        481       0.52      0.273      0.311      0.211

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/25      2.33G      1.191        1.3      1.426         26        640: 100% ━━━━━━━━━━━━ 240/240 5.9it/s 40.4s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 5.1it/s 1.8s0.2s
                   all        259        481      0.807      0.514       0.58      0.398

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/25      2.33G       1.13      1.208      1.386         27        640: 100% ━━━━━━━━━━━━ 240/240 5.5it/s 43.7s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 4.3it/s 2.1s0.3s
                   all        259        481      0.613      0.616      0.642      0.447

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/25      2.35G      1.104      1.167      1.363         43        640: 100% ━━━━━━━━━━━━
```

```
━━━━━━ 240/240 5.6it/s 42.6s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.6it/s 1.9s0.3s
                   all        259        481      0.703      0.648      0.686       0.49

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/25      2.35G      1.076       1.12       1.34         31        640: 100% ━━━━━━
━━━━━━ 240/240 5.4it/s 44.4s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 3.8it/s 2.4s0.3s
                   all        259        481      0.704      0.319       0.42       0.29

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      10/25      2.35G      1.036      1.069      1.313         32        640: 100% ━━━━━━
━━━━━━ 240/240 5.6it/s 42.7s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.4it/s 2.0s0.3s
                   all        259        481      0.724      0.709      0.759      0.545

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/25      2.35G      1.049      1.054      1.323         38        640: 100% ━━━━━━
━━━━━━ 240/240 5.7it/s 41.9s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.1it/s 2.2s0.3s
                   all        259        481      0.877      0.591      0.722      0.513

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/25      2.35G      1.007     0.9911      1.291         43        640: 100% ━━━━━━
━━━━━━ 240/240 6.0it/s 40.0s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.4it/s 1.7s0.2s
                   all        259        481       0.89      0.624      0.778      0.564

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/25      2.35G      0.981     0.9581      1.273         41        640: 100% ━━━━━━
━━━━━━ 240/240 6.2it/s 38.7s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.3it/s 1.7s0.2s
                   all        259        481      0.809      0.756        0.8      0.593

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/25      2.35G     0.9683     0.9423      1.268         35        640: 100% ━━━━━━
━━━━━━ 240/240 5.7it/s 42.0s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 3.8it/s 2.4s0.3s
                   all        259        481      0.741      0.706      0.758      0.579

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/25      2.36G     0.9542     0.9073      1.264         30        640: 100% ━━━━━━
━━━━━━ 240/240 5.5it/s 43.8s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.2it/s 1.7s0.2s
                   all        259        481      0.777      0.763       0.81      0.604
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/25      2.36G     0.9571     0.7981      1.263         20        640: 100% ━━━━━━
━━━━━━ 240/240 5.9it/s 40.9s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.6it/s 2.0s0.3s
```

```
                 all        259        481      0.847      0.684      0.785      0.604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/25      2.36G      0.929     0.7494      1.251         17        640: 100% ━━━━━━
━━━━━ 240/240 5.6it/s 43.1s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.0it/s 2.2s0.3s
                 all        259        481      0.806      0.784      0.825      0.618

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/25      2.36G     0.9076     0.7024       1.23         22        640: 100% ━━━━━━
━━━━━ 240/240 5.6it/s 43.0s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.3it/s 1.7s0.2s
                 all        259        481      0.808      0.808      0.822      0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/25      2.36G     0.8795     0.6745       1.21         17        640: 100% ━━━━━━
━━━━━ 240/240 5.9it/s 40.5s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.8it/s 1.9s0.2s
                 all        259        481      0.839      0.781      0.817      0.601

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/25      2.36G     0.8644     0.6531      1.193         16        640: 100% ━━━━━━
━━━━━ 240/240 6.1it/s 39.2s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.2it/s 1.7s0.2s
                 all        259        481      0.855      0.809      0.849      0.636

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      21/25      2.36G     0.8543     0.6269      1.182         16        640: 100% ━━━━━━
━━━━━ 240/240 5.8it/s 41.2s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.1it/s 2.2s0.3s
                 all        259        481      0.869      0.823      0.855      0.645

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      22/25      2.36G     0.8205      0.604      1.158         16        640: 100% ━━━━━━
━━━━━ 240/240 5.4it/s 44.1s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 4.3it/s 2.1s0.3s
                 all        259        481      0.791      0.819      0.842      0.641

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      23/25      2.36G     0.8068     0.5608       1.15         16        640: 100% ━━━━━━
━━━━━ 240/240 5.6it/s 42.6s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.2it/s 1.7s0.2s
                 all        259        481      0.855      0.819      0.863      0.673

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      24/25      2.38G     0.7955     0.5496       1.14         19        640: 100% ━━━━━━
━━━━━ 240/240 6.2it/s 38.8s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 9/9 5.0it/s 1.8s0.2s
                 all        259        481      0.869      0.834      0.865      0.681

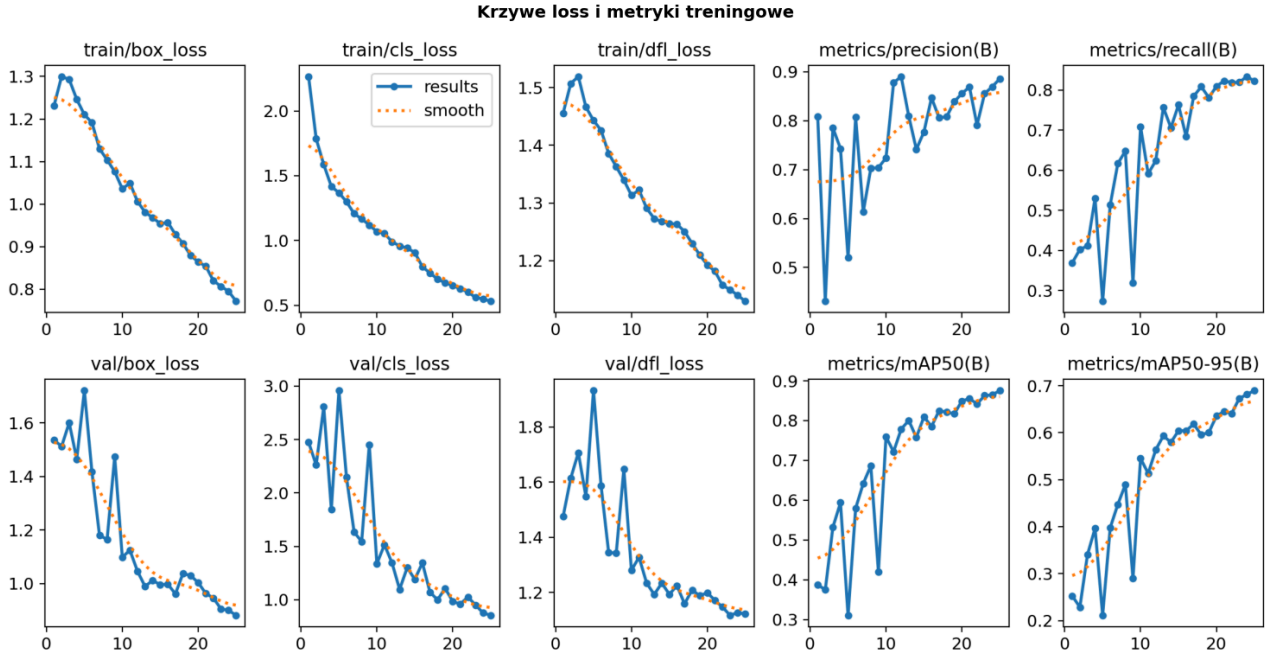
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      25/25      2.38G     0.7719     0.5318       1.13         19        640: 100% ━━━━━━
```

```
━━━━━━━━━━━━ 240/240 5.6it/s 42.6s<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━━━━━ 9/9 5.1it/s 1.8s0.2s
                   all        259        481      0.886      0.822      0.876       0.69

25 epochs completed in 0.316 hours.
Optimizer stripped from C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs
\detect\weapon_detector\weights\last.pt, 6.2MB
Optimizer stripped from C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs
\detect\weapon_detector\weights\best.pt, 6.2MB

Validating C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\detect\weapo
n_detector\weights\best.pt...
Ultralytics 8.4.14  Python-3.12.6 torch-2.6.0+cu124 CUDA:0 (NVIDIA GeForce RTX 3060 Laptop
GPU, 6144MiB)
Model summary (fused): 73 layers, 3,006,038 parameters, 0 gradients, 8.1 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━━━━━ 9/9 3.3it/s 2.7s0.3s
                   all        259        481      0.886      0.822      0.875       0.69
                  guns        161        306      0.866       0.65      0.777      0.588
                 knife         93        175      0.905      0.994      0.973      0.793
Speed: 0.3ms preprocess, 2.2ms inference, 0.0ms loss, 2.0ms postprocess per image
Results saved to C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\detect
\weapon_detector
MLflow: results logged to runs\mlflow
MLflow: disable with 'yolo settings mlflow=False'

Katalog wyników: runs/detect\weapon_detector\
```

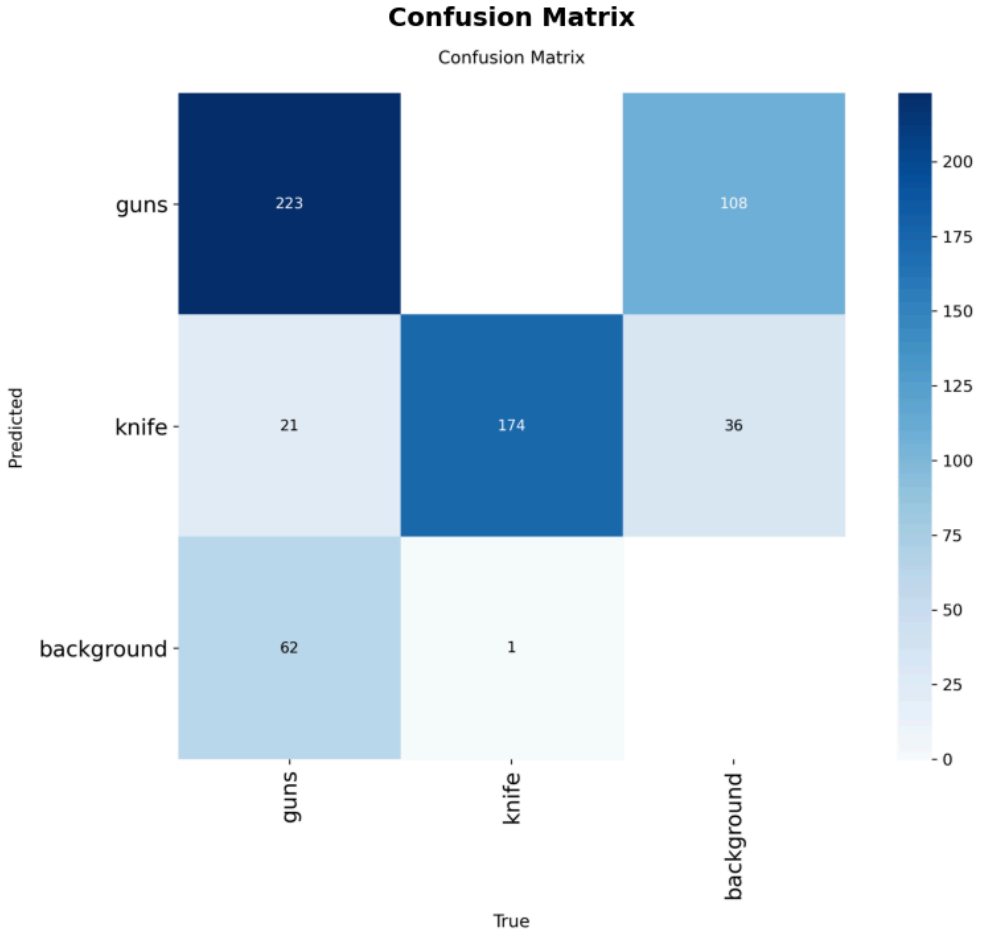


3. Ewaluacja modelu

Ocena najlepszego modelu na zbiorze testowym/walidacyjnym. Wyświetlenie metryk mAP50, mAP50-95, Precision, Recall oraz Confusion Matrix.

```
Ładowanie najlepszego modelu: runs/detect\weapon_detector\weights\best.pt
Ultralytics 8.4.14  Python-3.12.6 torch-2.6.0+cu124 CUDA:0 (NVIDIA GeForce RTX 3060 Laptop
GPU, 6144MiB)
Model summary (fused): 73 layers, 3,006,038 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access  (ping: 0.10.0 ms, read: 76.332.5 MB/s, size: 26.3 KB)
val: Scanning C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\data\test\labe
ls.cache... 259 images, 5 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 259/259  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
100% ━━━━━━━━━━━━ 17/17 4.7it/s 3.6s0.1s
                   all        259        481      0.887      0.822      0.875       0.69
                  guns        161        306      0.868       0.65      0.776      0.588
                 knife         93        175      0.905      0.994      0.974      0.793
Speed: 2.6ms preprocess, 4.6ms inference, 0.0ms loss, 2.0ms postprocess per image
Results saved to C:\github\glebokie-uczenie-praca-zal\weapon-detector-notebook\runs\detect
\val

==================================================
WYNIKI EWALUACJI
==================================================
mAP50:     0.8750
mAP50-95:  0.6903
Precision: 0.8867
Recall:    0.8223
```

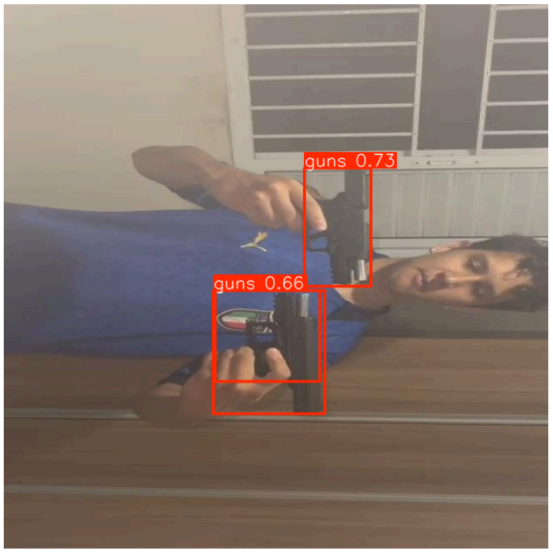


4. Inferencja na nowych zdjęciach

Test modelu na zdjęciach z broni (ze zbioru testowego) oraz zdjęciach **bez broni** (pobrane z internetu) — weryfikacja czy model poprawnie odróżnia broń od nie-broni.

Inferencja: broń vs nie-broń

Broń (test set)
Detekcje (3): guns: 0.73, guns: 0.66, guns: 0.41



Broń (test set)
Detekcje (2): knife: 0.78, knife: 0.56

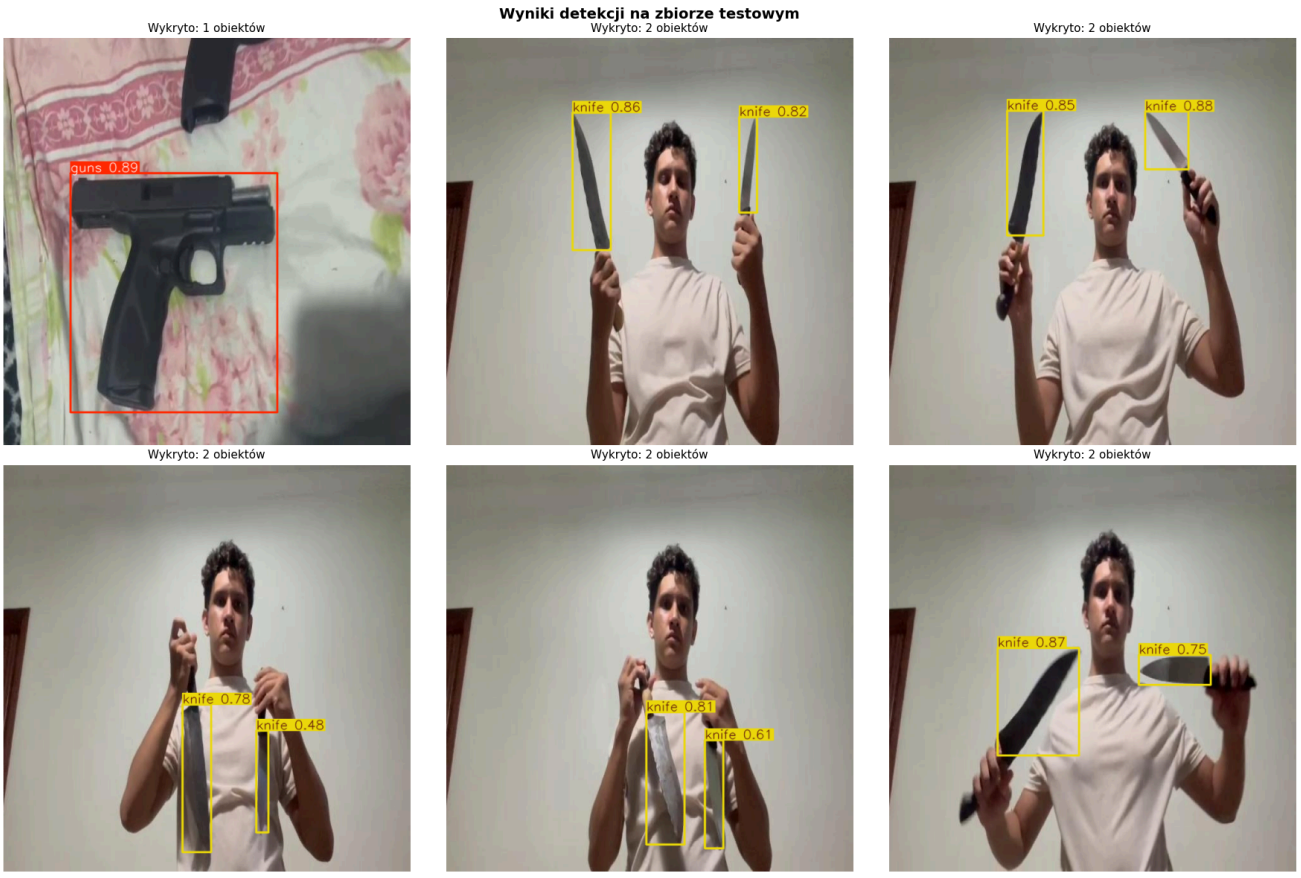


Nie-broń (internet)
Detekcje (0): Brak detekcji



Nie-broń (internet)
Detekcje (0): Brak detekcji





5. Podsumowanie

Model YOLOv8n został wytrenowany na datasecie Weapon-2 (Roboflow) do detekcji broni (`guns`) i noży (`knife`). Poniżej podsumowanie wyników.

```
============================================================
PODSUMOWANIE PROJEKTU: Gun Detector
============================================================
Model:        YOLOv8n (fine-tuned)
Dataset:      Weapon-2 (Roboflow, CC BY 4.0)
Klasy:        ['guns', 'knife']
Train imgs:   3839
Test imgs:    259
Epochs:       25
mAP50:        0.8750
mAP50-95:     0.6903
Precision:    0.8867
Recall:       0.8223

Best model:   runs/detect\weapon_detector\weights\best.pt
Screenshots: images/
============================================================
```